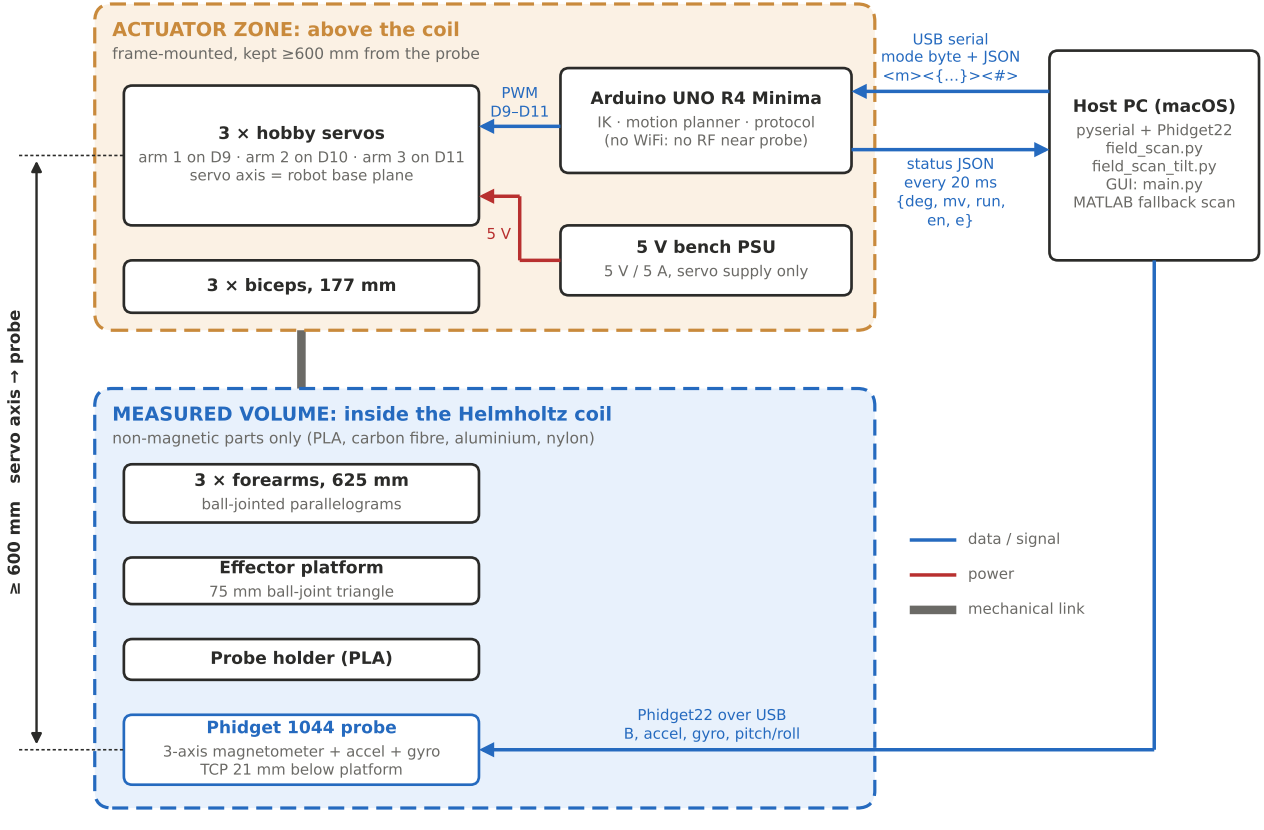
ACTUATOR ZONE: above the coil
frame-mounted, kept ≥600 mm from the probe
3 × hobby servos
arm 1 on D9 · arm 2 on D10 · arm 3 on D11
servo axis = robot base plane
PWM
D9–D11
Arduino UNO R4 Minima
IK · motion planner · protocol
(no WiFi: no RF near probe)
5 V
5 V bench PSU
5 V / 5 A, servo supply only
3 × biceps, 177 mm
USB serial
mode byte + JSON
<m><{...}><#>
status JSON
every 20 ms
{deg, mv, run,
en, e}
Host PC (macOS)
pyserial + Phidget22
field_scan.py
field_scan_tilt.py
GUI: main.py
MATLAB fallback scan
≥ 600 mm servo axis → probe
MEASURED VOLUME: inside the Helmholtz coil
non-magnetic parts only (PLA, carbon fibre, aluminium, nylon)
3 × forearms, 625 mm
ball-jointed parallelograms
Effector platform
75 mm ball-joint triangle
Probe holder (PLA)
Phidget 1044 probe
3-axis magnetometer + accel + gyro
TCP 21 mm below platform
Phidget22 over USB
B, accel, gyro, pitch/roll
data / signal
power
mechanical link